

JS SYLLABUS



Presented by Livesh Garg

Week 1 – JavaScript Fundamentals

Class 1: Introduction to JavaScript

- **What is JavaScript?**
- **How JavaScript works in browsers**
- **Adding JavaScript to HTML**
- **Variables (var, let, const)**
- **Data Types**
 - **String**
 - **Number**
 - **Boolean**
 - **Null**
 - **Undefined**

- **Comments**

Practice

- **Calculator for two numbers**
- **User information display**

Week 1 – JavaScript Fundamentals

Class 2: Operators & Conditions

- Arithmetic Operators
- Comparison Operators
- Logical Operators
- Type Conversion
- if
- if-else
- else-if

- Ternary Operator
- switch

Practice

- Grade Calculator
- Age Eligibility Checker

Week 1 – JavaScript Fundamentals

Class 3: Loops & Functions

- **for**
- **while**
- **do-while**
- **Break & Continue**
- **Functions**
- **Parameters**
- **Return Values**
- **Arrow Functions**

Practice

- **Multiplication Table**
- **Number Guessing Logic**
- **Basic Calculator Functions**

Week 2 – Arrays, Objects & DOM

Class 4: Arrays

- **Creating Arrays**
- **Accessing Elements**
- **Array Methods**
 - **push**
 - **pop**
 - **shift**
 - **unshift**
 - **splice**
 - **slice**
- **Looping Arrays**

Practice

- **Student Marks System**
- **To-Do List Data**

Week 2 – Arrays, Objects & DOM

Class 5: Objects

- **Object Creation**
- **Properties & Methods**
- **Nested Objects**
- **Object Destructuring**
- **Spread Operator**

Practice

- **Student Management System**
- **Product Catalog**

Week 2 – Arrays, Objects & DOM

Class 6: DOM Manipulation

- **DOM Introduction**
- **Selecting Elements**
- **Changing Content**
- **Changing Styles**
- **Creating Elements**
- **Removing Elements**
- **Event Listeners**

Practice

- **Dynamic To-Do App**
- **Theme Changer**

Week 3 – Modern JavaScript (ES6+)

Class 7: Advanced Functions

- **Callback Functions**
- **Higher Order Functions**
- **Closures**
- **Scope**
- **Hoisting**

Practice

- **Counter App**
- **Custom Callback Examples**

Week 3 – Modern JavaScript (ES6+)

Class 8: Array Advanced Methods

- **map()**
- **filter()**
- **reduce()**
- **find()**
- **some()**
- **every()**
- **sort()**

Practice

- **Student Result Dashboard**
- **Product Filtering**

Week 3 – Modern JavaScript (ES6+)

Class 9: Async JavaScript

- Synchronous vs Asynchronous
- `setTimeout`
- `setInterval`
- Callbacks
- Promises
- `async/await`

Practice

- Loading Simulation
- API Fetch Example

Week 4 – Advanced JavaScript & Real Projects

Class 10: APIs & JSON

- JSON
- Fetch API
- GET Request
- POST Request
- Error Handling

Practice

- Weather App
- User Data Fetcher

Week 4 – Advanced JavaScript & Real Projects

Class 11: OOP & Modules

- **Classes**
- **Objects**
- **Constructor**
- **Inheritance**
- **Encapsulation**
- **Import / Export Modules**

Practice

- **Student Class System**
- **Product Management System**

Week 4 – Advanced JavaScript & Real Projects

Class 12: Final Projects & Interview Preparation

- **Local Storage**
- **Session Storage**
- **Debugging**
- **Browser DevTools**
- **JavaScript Interview Questions**
- **Best Practices**

Practice

- **To-Do Application**
- **Quiz Application**
- **Weather Application**
- **Expense Tracker**
- **Notes App**
- **E-Commerce Product Filter**

Final Outcome

After 12 classes, students will know:

- ✓ **JavaScript Fundamentals**
- ✓ **Arrays & Objects**
- ✓ **DOM Manipulation**
- ✓ **ES6 Features**
- ✓ **Async JavaScript**

- ✓ **APIs & JSON**
- ✓ **OOP Concepts**
- ✓ **Local Storage**
- ✓ **Real-World Projects**
- ✓ **Interview Preparation**

Contact Us

FOR FURTHER INQUIRIES, FEEL FREE TO
REACH OUT.



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Thank You!

WE APPRECIATE YOUR TIME.